

Wafer Yield Root Cause & Defect Pattern Analyser

AI-Assisted Semiconductor Manufacturing Analytics

Team CodeCraft — Track: AI/ML

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Wafer Yield Root Cause & Defect Pattern Analyser

AI-Assisted Semiconductor Manufacturing Analytics



The Challenge: Finding Why Yield Drops

Manufacturing scale

Wafer lot data • Equipment sensor streams • Defect images • Process parameters

Problem

Too many signals → hard to correlate abnormal conditions with yield loss and defects

Impact

Slow root-cause investigations → delayed corrective actions → sustained yield loss

Why it matters

Faster pattern detection enables earlier investigations and corrective measures

Our Solution: From Data to Engineering Insight

The Wafer Yield Root Cause & Defect Pattern Analyser is an AI-assisted platform that fuses lot, sensor, equipment and defect data to surface the patterns and evidence engineers need.

Detect patterns

Process & defect pattern
discovery

Risk prediction

Estimate yield degradation
risk

Root-cause evidence

Data-grounded explanations
for engineers

Corrective actions

Actionable engineering
recommendations

Lot & fleet views

From single-wafer to fleet trends

Ask Bob

Conversational engineering assistance

How the System Works



Diagram shows data flow from synthetic dataset through processing, analytics engine and engineer-facing outputs with Ask Bob integration and local fallback.

AI-Powered Yield Risk Prediction

ANALYTICS

Lot Overview

Root Cause Analysis

Risk Prediction

Ask Bob

IBM WaferSight

Wafer Yield Root Cause & Defect Pattern Analyser

watsonx.ai OFFLINE

Batch Risk Prediction

Random Forest model trained on 500 historical lots — enter upcoming lot parameters to forecast yield risk.

QUICK TEST SCENARIOS

ETCH-03 hot run (high risk)

DEP-02 RF spike

High particle contamination

Normal conditions

LOT CONFIGURATION

NODE

PROCESS STEP

TOOL

RECIPE

3nm

etch

ETCH-01

R-ETH-04

SENSOR OVERRIDES (optional — leave blank to use fleet averages)

ETCH TEMP (°C)

RF POWER (W)

MEAN DEFECT DENSITY (/CM²)

191

425

1

Predict Risk

Try: ETCH-03 + etch temp 191°C for a high-risk example

PREDICTION RESULT

52%

HIGH RISK

probability of low yield

TOP CONTRIBUTING FACTORS

↓ particle_density_mean

24.2%

↓ defect_density_max

23.9%

↓ defect_density_mean

22.3%

↓ defect_count

14.1%

↓ particle_count

8.3%

↓ alert_readings_count

2.1%

↓ etch_temp_c

0.8%

↓ etch_pressure_mtorr

0.7%

HIGH: High probability of low yield. Immediate review recommended before scheduling this lot.

Flow

1. Configure Lot

Node • Process Step • Tool • Recipe

2. Enter Process Conditions

Etch Temperature • RF Power • Defect Density

3. Random Forest Analysis

Analyzes historical wafer-lot patterns

4. Predict Yield Risk

LOW • MEDIUM • HIGH

5. Identify Contributing Factors

Highlights parameters affecting the predicted risk

6. Support Engineering Action

Enables faster investigation and corrective action

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Built with IBM AI Technology



IBM watsonx.ai

Used for AI-powered conversational analysis when configured; integrates with Ask Bob for contextual queries.



Granite (Generative)

Generative capability for Ask Bob. We use grounded prompts and a local fallback for reliability.



Local grounded fallback

Ensures the assistant remains usable if external services are unavailable.

Supporting stack: Python • FastAPI • pandas • NumPy • scikit-learn (Random Forest) • React

Results & Impact

500 Historical Wafer Lots Analyzed



Faster investigation

Accelerated triage reveals suspicious yield and defect patterns earlier in the analysis cycle



Early risk detection

Random Forest model classifies wafer lots into Low / Medium / High risk to prioritise engineering attention.



Evidence-based root cause

Model highlights process steps, equipment, sensor signals, and defect signatures correlated with failures.



Actionable insights

Clear corrective-action recommendations for engineers to investigate and mitigate identified risks.

Operational flow

Manufacturing Data → Pattern Detection → Risk Prediction → Root Cause → Corrective Action

Future deployment

Planned data sources to transition from prototype to production integration.

Synthetic Data

Used for model development and scenario testing.

Live MES / SCADA / Equipment Data

Target for operational deployment and continuous monitoring.